General Mathematics — Exercise Sheet 30

Uniform convergence and power series

Variant: Statements

Exercise 1 (A large error moving towards zero). For $n \geq 1$, define $f_n(x) = nx/(1+n^2x^2)$ on $[0,1]$. Find its pointwise limit and determine whether convergence is uniform on $[0,1]$ and on $[\delta,1]$ for a fixed $0 < \delta \leq 1$. Use the quantifier definitions from Lecture 30 and give an explicit moving point whenever uniform convergence fails.

Exercise 2 (A uniform series on an unbounded domain). Consider the trigonometric series

$$F(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2+1}, \qquad x \in \mathbb{R}.$$

Use Lecture 30's majorant test to prove absolute and uniform convergence on $\mathbb{R}$, and continuity of F there. Express $\int_0^{\pi} F(x)\,\mathrm{d}x$ as a numerical series involving only odd indices. Justify every exchange of an infinite sum and an integral.

Exercise 3 (Radii, endpoints and the whole convergence interval). Determine the radius and full interval of convergence of both series:

$$\text{(a)} \quad \sum_{n=1}^{\infty} \frac{n^2(x+1)^{2n}}{4^n}, \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{(x-3)^n}{5^n n(n+1)}.$$

For each, decide whether convergence is uniform on its entire interval of convergence. For (b), compute the exact largest error after retaining indices 1 through N, over the full interval. Use Lecture 29's root and telescoping results and Lecture 30's majorant test. You may first prove that uniform convergence of a series forces its individual summands to tend uniformly to zero.

Exercise 4 (Dirichlet's test from finite summation). Let (a_n) be real, and let $A_0 = 0$ and $A_n = \sum_{j=1}^{n} a_j$. Suppose $|A_n| \leq K$ for all n, where $K \geq 0$. Let $b_n \geq 0$ be nonincreasing with $b_n \to 0$.

(a) Prove the finite identity, for integers $M > N \geq 0$,

$$\sum_{n=N+1}^{M} a_n b_n = A_M b_M - A_N b_{N+1} + \sum_{n=N+1}^{M-1} A_n(b_n - b_{n+1}),$$

where an empty sum is zero. Deduce a tail bound $2Kb_{N+1}$ and use Lecture 29's Cauchy criterion to prove convergence of $\sum a_n b_n$.

(b) Suppose a_n are instead real functions on a nonempty set $E \subseteq \mathbb{R}$, with their partial sums bounded by the same K at every point. Keep b_n independent of the point. Prove uniform convergence and the uniform remainder bound $2Kb_{N+1}$.

(c) Apply these results to $a_n(\theta) = \sin(n\theta)$ and $b_n = 1/n$. Prove pointwise convergence for every $\theta \in \mathbb{R}$, and uniform convergence on $[\delta, 2\pi - \delta]$ for every $0 < \delta < \pi$. Use Lecture 16's complex exponential product law and Lecture 29's finite geometric identity.

Exercise 5 (Arctangent by integration and an endpoint limit). Use Lecture 30's uniform termwise integration to prove

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \qquad (-1 \le x \le 1).$$

First work strictly inside $(-1, 1)$, then justify both endpoints. Prove that the series is uniform on $[-1, 1]$, with error after indices 0 through N at most $1/(2N+3)$. Finally use five terms at $x = 1/2$ to give a rational approximation with error less than $1/10000$, including the sign of the error. Recall Lecture 29's geometric sum and alternating error bound, Lecture 8's arctangent derivative, and Lecture 26's endpoint evaluation.